

QUALIFICATION FILE

Junior software Developer

- ☒ Short Term Training (STT) ☐ Long Term Training (LTT) ☐ Apprenticeship
☒ Upskilling ☐ Dual/Flexi Qualification ☐ For ToT
☐ For ToA
- ☒ General ☐ Multi-skill (MS) ☐ Cross Sectoral (CS) ☐ Future Skill ☐ OEM

NCrF/NSQF Level: 3

Submitted By: Namrata Kapur

IT-ITeS Sector Skills Council NASSCOM (SSC NASSCOM)

Plot No. – 7, 8, 9 & 10

Sector – 126, Noida, Uttar Pradesh - 201303

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Section 1: Basic Details

1.	Qualification Name	Junior Software Developer	
2.	Sector/s	IT/ITeS	
3.	Type of Qualification: ☐ New ☒ Revised ☐ Has Electives/Options ☐ OEM	NQR Code & version of the existing /previous qualification: 2022/ITES/ITSSC/05247 & Version 2	Qualification Name of the existing/previous version: Junior software Developer
4.	Qualification Name (Wherever applicable)	Junior Software Developer	
5.	National Qualification Register (NQR) Code &Version (Will be issued after NSQC approval)	QG-03-IT-01582-2023-V1.1-NASSCOM & Version 3	6. NCrF/NSQF Level: 3
7.	Award (Certificate/Diploma/Advance Diploma/ Any Other (Wherever applicable specify multiple entry/exits also & provide details in annexure)	Certificate	
8.	Brief Description of the Qualification	Individuals in this job are assigned one of the many entry level roles in the software industry including support and help desk, testing, user interaction design, maintenance, enhancement, development, and documentation. They are responsible for assisting in performing the key activities and tasks involved in the assigned role.	
9.	Eligibility Criteria for Entry for a Student/Trainee/Learner/Employee	Entry Qualification & Relevant Experience: *Relevant Experience: Computer Operation The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications. Grade 10 OR 8th Grade Pass with 2-year relevant experience* OR Previous Relevant qualification of NSQF level 2 with 1 years of relevant experience* Min Age: 14 Years	
10.	Credits Assigned to this Qualification, Subject to Assessment (as per National Credit Framework (NCrF))	15 Credits	11. Common Cost Norm Category (I/II/III) (wherever applicable): II

12.	Any Licensing Requirements for Undertaking Training on This Qualification (wherever applicable)	NA																							
13.	Training Duration by Modes of Training Delivery (Specify Total Duration as per selected training delivery modes and as per requirement of the qualification)	☒ Offline Only ☒ Online Only ☐ Blended																							
		<table border="1"> <thead> <tr> <th>Training Delivery Mode</th> <th>Theory (Hours)</th> <th>Practical (Hours)</th> <th>OJT (Mandatory) Hours</th> <th>OJT (Recommended) Hours</th> <th>Total (Hours)</th> </tr> </thead> <tbody> <tr> <td>Classroom (offline)</td> <td>110</td> <td>280</td> <td>60</td> <td>-</td> <td>450</td> </tr> <tr> <td>Online</td> <td>110</td> <td>280</td> <td>60</td> <td>-</td> <td>450</td> </tr> </tbody> </table>						Training Delivery Mode	Theory (Hours)	Practical (Hours)	OJT (Mandatory) Hours	OJT (Recommended) Hours	Total (Hours)	Classroom (offline)	110	280	60	-	450	Online	110	280	60	-	450
Training Delivery Mode	Theory (Hours)	Practical (Hours)	OJT (Mandatory) Hours	OJT (Recommended) Hours	Total (Hours)																				
Classroom (offline)	110	280	60	-	450																				
Online	110	280	60	-	450																				
		(Refer Blended Learning Annexure for details)																							
14.	Aligned to NCO/ISCO Code/s (if no code is available mention the same)	NCO-2015/ 2512.0205																							
15.	Progression Path After Attaining the Qualification, wherever applicable (Please show Professional and Academic progression)	This entry should refer to one or more of the following: Professional progression: access to related qualification(s) at the next NSQF level: Assistant Software Developer																							
16.	Other Indian languages in which the Qualification & Model Curriculum are being submitted	Hindi																							
17.	Is similar Qualification(s) available on NQR-if yes, justification for this qualification	☐ Yes ☒ No URLs of similar Qualifications:																							
18.	Is the Job Amenable to Persons with Disability	☒ Yes ☐ No If "Yes", specify applicable type of Disability: Visual, Hearing or Speech impairment, Locomotor Disability																							
19.	How will participation of women be encouraged?	The Program is gender neutral although to increase women's participation, organizations are keeping aside a few seats to encourage female candidates.																							
20.	Are Greening/Environment Sustainability Aspects covered (Specify the NOS/Module which Covers it)	☐ Yes ☒ No																							

21.	Is Qualification suitable to be offered in Schools/Colleges	Schools: ☐ Yes ☒ No	Colleges ☒ Yes ☐ No
22.	Name and Contact Details Submitting / Awarding Body SPOC (In case of CS or MS, provide details of both Lead AB & Supporting ABs)	Name: Namrata Kapur Email: Namrata@nasscom.in Contact No.: 0120-4990111 Website: https://nasscom.in	
23.	Final Approval Date by NSQC: 27 th January 2022	24. Validity Duration: 3 Years	25. Next Review Date: 27 th January 2025

Section 2: Module Summary

NOS/s of Qualification

(In Exceptional cases these could be described as components)

Mandatory NOS/s:

Specify the training duration and assessment criteria at NOS/Module level. For Further details refer curriculum document.

Th.-Theory **Pr.**-Practical **OJT**-On the Job training **Man.**-Mandatory Training **Rec.**-Recommended **Proj.**- Project

S.No.	NOS Module Name	NOS/Module Code & Version (If Applicable)	Core/Non-Core	NCrF/NSQF Level	Credits as per NcrF	Training Duration (Hours)					Assessment Marks					
						Th.	Pr.	OJT-Man.	OJT-Rec.	Total	Th.	Pr.	Proj.	Viva	Total	Weightage (%) (if applicable)
1.	Assist in performing software construction and software testing entry-level tasks in the IT Services industry.	SSC/N0506 V2.0	Core	3	11	86:00	244:00	00:00	00:00	330:00	30	70	-	-	100	70%
2.	Employability Skills (60 Hours)	DGT/VSQ/N0102, V1.0	Non-Core	4	02	24:00	36:00	00:00	00:00	60:00	20	30	-	-	50	30%
3.	OJT	-	-	-	02	00:00	00:00	60:00	00:00	60:00						
Duration (in Hours)/Total Marks					15	110:00	280:00	60:00	00:00	450:00	50	100	-	-	150	100%

Assessment - Minimum Qualifying Percentage

Please specify **any one** of the following:

Minimum Pass Percentage – Aggregate at qualification level: 50 % (Every Trainee should score specified minimum aggregate passing percentage at qualification level to successfully clear the assessment.)

Section 3: Training Related

1.	Trainer's Qualification and experience in the relevant sector (in years) (as per NCVET guidelines)	Educational Qualification: Minimum 12 th Standard. Industry experience: Minimum 2 years' experience in Software development domain Training experience: Minimum 1-year experience Certification: "Trainer" mapped to the Qualification Pack "MEP/Q2601, V2.0" Minimum accepted score is 80% aggregate.
2.	Master Trainer's Qualification and experience in the relevant sector (in years) (as per NCVET guidelines)	Educational Qualification: Minimum 12 th Standard. Industry experience: Minimum 2 years' experience in Software development domain Training experience: Minimum 1-year experience Certification: "Master Trainer" mapped to the Qualification Pack "MEP/Q2602, V2.0" Minimum accepted score is 80% aggregate.
3.	Tools and Equipment Required for the Training	☒ Yes ☐ No (If "Yes", details to be provided in Annexure)
4.	In Case of Revised Qualification, details of Any Upskilling Required for Trainer	NA

Section 4: Assessment Related

1.	Assessor's Qualification and experience in relevant sector (in years) <i>(as per NCVET guidelines)</i>	Educational Qualification: Graduate in any discipline Industry experience: 2 years of experience in Software Development domain Training experience: 2 years Certification: Job role "Assessor (VET and SKILLS)" mapped to the Qualification Pack "MEP/Q2701, V2.0" with a minimum score of 80%
2.	Proctor's Qualification and experience in relevant sector (in years) <i>(as per NCVET guidelines), (wherever applicable)</i>	Educational Qualification: Graduate in any discipline Industry experience: 2 years of experience in Software Development domain Training experience: 2 years Certification: Job role "Assessor (VET and SKILLS)" mapped to the Qualification Pack "MEP/Q2701, V2.0" with a minimum score of 80%
3.	Lead Assessor's/Proctor's Qualification and experience in relevant sector (in years) <i>(as per NCVET guidelines)</i>	Educational Qualification: Graduate in any discipline Industry experience: 2 years of experience in Software Development domain Training experience: 2 years Certification: Job role "Assessor (VET and SKILLS)" mapped to the Qualification Pack "MEP/Q2701, V2.0" with a minimum score of 80%
4.	Assessment Mode <i>(Specify the assessment mode)</i>	Can be either in the classroom or online
5.	Tools and Equipment Required for Assessment	☒ Same as for training ☐ Yes ☐ No <i>(details to be provided in Annexure-if it is different for Assessment)</i>

Section 5: Evidence of the Need for the Qualification

Provide Annexure/Supporting documents name.

1.	Latest Skill Gap study (not older than 2 years) (Yes/No): Yes
2.	Latest Market Research Reports or any other source (not older than 2 years) (Yes/No): Yes
3.	Government/Industry initiatives/requirement (Yes/No): Yes
4.	Number of industry validations provided:
5.	Estimated number of people to be trained and employed: Yes
6.	Evidence of Concurrence/Consultation with Line/State Departments: Yes If “No”, why:

Section 6: Annexure & Supporting Documents Check List

Specify Annexure Name / Supporting document file name

1.	Annexure: NCrf/NSQF level justification based on NCrf/NSQF descriptors (<i>Mandatory</i>)	Evidence of Level
2.	Annexure: List of tools and equipment relevant for NOS (<i>Mandatory, except in case of online course</i>)	Tools and Equipment (lab set-up)
3.	Annexure: Detailed Assessment criteria (<i>Mandatory</i>)	Performance Criteria Details
4.	Annexure: Assessment Strategy (<i>Mandatory</i>)	Assessment Strategy
5.	Annexure: Blended Learning (<i>Mandatory, in case selected Mode of delivery is Blended Learning</i>)	NA
6.	Annexure: Multiple Entry Exit Details (<i>Mandatory, in case qualification has multiple entry-exit</i>)	NA
7.	Annexure: Acronym and Glossary (<i>Optional</i>)	NA
8.	Supporting Document: Model Curriculum (<i>Mandatory-Public View</i>)	MC_SSCQ0508-Junior software Developer_V3.0
9.	Supporting Document: Career Progression (<i>Mandatory-Public View</i>)	Occupational Map – ITS

10.	Supporting Document: Occupational Map (<i>Mandatory</i>)	Occupational Map - ITS
11.	Supporting Document: Assessment SOP (<i>Mandatory</i>)	Assessment Strategy
12.	Any Other document you wish to submit:	NA

Annexure: Evidence of Level

NCrF/NSQF Level Descriptors	Key requirements of the job role/ outcome of the qualification	How the job role/ outcomes relate to the NCrF/NSQF level descriptor	NCrF/NSQF Level
Professional Theoretical Knowledge/Process knowledge	- Assist in performing software construction and software testing entry-level tasks in the IT Services Industry - Manage your work to meet requirements - Work effectively with colleagues - Maintain a healthy, safe, and secure working environment - Provide data/information in standard formats	Individual in this job needs to design algorithms to solve problems and convert them into code using the appropriate programming language constructs. Also, he/she needs to read and execute a test case and record the outcome in the appropriate template. The above-mentioned algorithm-based tasks are highly process oriented in nature & he/she must work in limited range & predictable situations but leading to clear logical choices.	3
Professional and Technical Skills/ Expertise/ Professional Knowledge	- The responsibilities, policies and guidelines associated with entry level roles for software developers - The formal and informal continuous learning opportunities offered by the organization - How to access and update organization's knowledge base - The scope of work to be carried out and the importance of keeping within the boundaries	Individual in this job needs to have basic knowledge required to design algorithms to solve problems and convert them into code using the appropriate programming language constructs. He/she needs to have the basic knowledge of the field of application development, such as programming language fundamentals, implementation of algorithms, data base skills including DBMS, data design, and querying table structures for specific data, software engineering	3

NCrF/NSQF Level Descriptors	Key requirements of the job role/ outcome of the qualification	How the job role/ outcomes relate to the NCrF/NSQF level descriptor	NCrF/NSQF Level
	- Who to be involved in the process of collating feedback on the work output? - Standard templates and tools available and how to use them - Web browsers, the world wide web, internet mail, internet search, web based social media applications and web programming - Logical analysis, problem solving skills, process approach, algorithmic thinking, and pseudo code - Foundational computation and mathematics including binary arithmetic, number sense, discrete mathematics, and numerical descriptive measures of data - Fundamentals of programming language and implementation of algorithms - Fundamentals of databases including DBMS, data design, and querying table structures for specific data - Methods and key processes used to develop software applications - How to read a detailed program specification and implement it using a programming language - How to read a test case, execute the same and record the results of testing	approaches to develop applications and the key processes used for developing application software.	
Employment Readiness & Entrepreneurship Skills & Mind-set/Professional Skill	- Follow rule-based decision-making processes - Carry out rule-based transactions in line with customer-specific requirement - Apply problem-solving approaches to take decisions in different situations - Configure and analyse data to disseminate relevant information to others - Check if the work is complete and free from errors	Individual in this job needs to recall and demonstrate practical skills related to basic computer and internet literacy including operating a computer, describing its major components and how they work, using Windows and Linux OS, operating a browser, searching the internet, managing mails, and using social internet media.	3

NCrF/NSQF Level Descriptors	Key requirements of the job role/ outcome of the qualification	How the job role/ outcomes relate to the NCrF/NSQF level descriptor	NCrF/NSQF Level
	- Work effectively in a team environment with focus on individual and team strength - Use information technology effectively to input and/or extract data accurately	He/she needs to read and execute test cases in routine and repeat them in the narrow range of applications until required outcomes are met. The outcomes of these test cases are recorded using appropriate rules, tools & using quality concepts through appropriate templates.	
Broad Learning Outcomes/Core Skill	- Communicate with others in writing with attention to details - Follow instructions, guidelines, procedures, rules, and service level agreements - Listen effectively and orally communicate information accurately for clarification - Work effectively in a customer facing environment	The individual should possess good written language & communication skills, with the required clarity, to be able to communicate effectively with appropriate people w.r.t. assigned roles in– both oral and in written mediums. He/she needs to have skill to basic arithmetic and algebraic principles, to demonstrate knowledge of the foundational mathematical concepts in computing.	3
Responsibility	- Work requirements on Information (qualitative and quantitative) - Algorithms and steps in problem solving - People with specific skill set, like Line manager/peers/ subject matter experts	Individuals in this job are assigned one of the many entry level roles in the software industry including support and help desk, testing, user interaction design, maintenance, enhancement, development, and documentation. They are responsible for assisting in performing the key activities and tasks involved in the assigned role under close supervision. These tasks will require the individual to take some responsibility for own work and learning. Since the individual is working under supervision (as needed at Level 3), this QF is justified to be pegged at Level 3.	3

Annexure: Tools and Equipment (lab set-up)

Batch Size:

S. No.	Tool / Equipment Name	Specification	Quantity for specified Batch size
1	PC/Laptop with internet	With Wifi (2MBPS Dedicated)	1 Unit per Trainee
2	Relevant Software: CRM Tool for demonstration	HTML, C++ / Java, IDE	1 Unit per Trainee
3	Microphone/Voice System	For lecture & class activities	1 Unit for Trainer
4	White Board		1 Unit for Trainer
5	White Board Maker		1 Unit for Trainer
6	Projector		1 Unit

Annexure: Industry Validations Summary

Provide summary information of all the industry validation in table. This is not required for OEM Qualifications.

S. No	Organisation Name	Representative Name	Designation	Contact Address	Contact Phone No	E-mail ID	LinkedIn Profile (if available)
1							
2							
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4							
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12							
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19							

Annexure: Training & Employment Details

Training & Employment Projections:

Year	Total Candidates		Women		People with Disability	
	Estimated Training #	Estimated Employed Opportunities	Estimated Training #	Estimated Employed Opportunities	Estimated Training #	Estimated Employed Opportunities
2023-24						
2024-25						
2025-26						

#The Estimated Data is an average for each state.

Training, Assessment, Certification, and Placement Data for previous versions of qualifications:

Qualification Version	Year	Total Candidates			Women			People with disability		
		Trained	Assessed	Certified	Trained	Assessed	Certified	Trained	Assessed	Certified
	2022-23									

*The total assessed candidate job role count is for 10 states.

List Schemes in which the previous version of qualification was implemented**Content availability for the previous version of qualifications:**

☐ Participant Handbook ☐ Facilitator Guide ☐ Digital Content ☐ Qualification Handbook ☐ Any Other:

Language in which content is available:**Annexure: Detailed Assessment Criteria**

Detailed Assessment criteria for each NOS/Module are as follows:

NOS/Module Name	Assessment Criteria for Performance Criteria/Learning Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
SSC/N0506: Assist in performing software construction and software testing entry-level tasks in the IT Services industry	PC1.operate a computer and describe the main components of a computer	-	10		
	PC2. be proficient on the use of various Browser operation, Internet search, Mails, and Social Interaction media	10	15		
	PC3. analyse information to reach a logical conclusion	5	15		
	PC4. design algorithms to solve problems and convert them into code using the appropriate programming language constructs.	10	20		
	PC5. read and execute a test case and record the outcome in the appropriate template.	5	5		
	PC6. communicate effectively with appropriate people w.r.t. assigned roles in simple English both oral and written.	-	5		
		30	70		
DGT/VSQ/N0102 Employability NOS for 60 Hours	PC1. Introduction to Employability Skills	1	1	-	-
	PC2. Constitutional values – Citizenship	1	1	-	-
	PC3. Becoming a Professional in the 21st Century	2	4	-	-
	PC4. Basic English Skills	2	3	-	-
	PC5. Career Development & Goal Setting	1	2	-	-
	PC6. Communication Skills	2	2	-	-
	PC7. Diversity & Inclusion	1	2	-	-
	PC8. Financial and Legal Literacy	2	3	-	-
	PC9. Essential Digital Skills	3	4	-	-
	PC10. Entrepreneurship	2	3	-	-
	PC11. Customer Service	1	2	-	-

NOS/Module Name	Assessment Criteria for Performance Criteria/Learning Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	PC12. Getting Ready for Apprenticeship & Jobs	2	3	-	-
	Total Marks	20	30	-	-
Grand Total Marks		50	100		

Annexure: Assessment Strategy

This section includes the processes involved in identifying, gathering, and interpreting information to evaluate the Candidate on the required competencies of the program.

Assessment System Overview

A uniform assessment of job candidates per industry standards facilitates the industry's progress by filtering employable individuals while simultaneously providing candidates with an analysis of personal strengths and weaknesses.

Assessment Criteria

The Sector Skill Council will create criteria for assessment for each Qualification Pack. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC. Assessment will be conducted for all compulsory NOS and where applicable, on the selected elective/option NOS/set of NOS.

Guidelines for Assessment			
Testing Environment	Tasks and Functions	Productivity	Teamwork
- Carry out assessments under realistic work pressures found in the normal industry workplace (or simulated workplace). - Ensure that the range of materials, equipment, and tools that learners use are current and of the type routinely found in the normal industry workplace (or simulated workplace) environments.	- Assess that all tasks and functions are completed in a way, and to a timescale that is acceptable in the normal industry workplace. - Assign workplace (or simulated workplace) responsibilities that enable learners to meet the requirements of the NOS.	Productivity levels must be checked to ensure that it reflects those that are found in the work situation being replicated.	Provide situations that allow learners to interact with the range of personnel and contractors found in the normal industry workplace (or simulated workplace).

Annexure: Acronym and Glossary

Acronym

Acronym	Description
AA	Assessment Agency
AB	Awarding Body
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On Job Training

Glossary

Term	Description
National Occupational Standards (NOS)	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process is obtained when a competent body determines that an individual has achieved learning outcomes to given standards.
Qualification File	A Qualification File is a template designed to capture necessary information about a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities based on their main economic function, product, service, or technology.

Annexure: Market Research & Gap Analysis

While collecting data from the companies for the occupational map, we also took feedback from industry, training institutions which was collected with respect to roles for which qualification files development, was to be prioritized. This was largely based on volume of people required, quantitative and qualitative shortfall which the industry feels they face. Governing council of IT - ITeS and Occupational Standards Committee which comprises of experts & senior leaders gave final approval and endorsement for the same.

The Indian IT industry's revenue is estimated to reach US\$ 194 billion in FY21, an increase of 2.3% YoY. The sector is the largest employer within the private sector. According to Gartner estimates, IT spending in India is estimated to reach US\$ 93 billion in 2021 (7.3% YoY growth) and further increase to US\$ 98.5 billion in 2022.

The IT industry accounted for 8% of India's GDP in 2020. Exports from the Indian IT industry are expected to increase by 1.9% to reach US\$ 150 billion in FY21. In 2020, the IT industry recorded 138,000 new hires. According to STPI (Software Technology Park of India), the software exports by its registered units increased by 7% YoY to reach Rs. 5 lakh crore (US\$ 67.40 billion) in FY21 from Rs. 4.66 lakh crore (US\$ 62.82 billion) in FY20, driven by rapid digitization and the IT industry's timely transition to remote working environments that helped to keep up the industry's growth amid coronavirus pandemic.

In Budget 2021, the government has allocated Rs. 53,108 crore (US\$ 7.31 billion) to the IT and telecom sector

In India there are 421 approved SEZs across the country, with 276 of them from IT & BPM and 145 as exporting SEZs. The IT & BPM industry's revenue is estimated at ~US\$ 194 billion in FY21, an increase of 2.3% YoY. The domestic revenue of the IT industry is estimated at US\$ 45 billion and export revenue is estimated at US\$ 150 billion in FY21.

Annexure: Government/Industry initiatives/requirement

Annexure: Evidence of Concurrence/Consultation with Line/State Departments: